

Spot Market Buy Order — Complete Test Steps

File: `tests/trade/spotMarketBuyOrder.spec.ts`

Data: `src/data/spotMarketBuyData.csv`

Mode: Serial (each step depends on the previous one)

SETUP — Before All Tests

Step 1: Launch a Chromium browser (non-headless, 1280×720 viewport) and open a new browser page.

Step 2: Navigate to the login page (`BASE_URL` + login route).

Step 3: Enter the email address in the email field and click "Continue".

Step 4: Enter the password in the password field and click "Continue".

Step 5: Read the current TOTP/OTP value and enter it in the OTP field, then click "Continue" to complete login successfully.

Step 6: Dismiss any post-login dialogs (popups, banners, announcements) and wait for the home/dashboard page to be fully loaded.

TC-01 — Navigate to Spot Trading

Step 7: Click on the Spot Trading link/menu item to navigate to the Spot Trading page and wait for it to load.

TC-02 — Verify Page Labels

Step 8: Read all visible labels on the Spot Trading page and verify each one is correct:

- "Depth View" label is present
- "Order Book" heading is present
- "Buy" tab is present
- "Sell" tab is present
- "Limit" tab is present
- "Market" tab is present
- "Market Trades" label is present
- "My Trades" label is present

- "All Orders" tab is present
 - "Trade History" label is present
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TC-03 — Search Currency Pair

Step 9: Type the pair name (e.g. `BTC/USDT` , read from CSV `searchPair` column) into the search/filter input in the market dropdown and wait for results to appear.

TC-04 — Mark as Favorite

Step 10: Click the star/favorite icon next to the pair in the search results to mark it as a favorite.

Step 11: Verify the success toast/status message confirms the pair was **added** to favorites.

TC-05 — Unmark from Favorites

Step 12: Click the star/favorite icon again to unmark the pair from favorites.

Step 13: Verify the status message confirms the pair was **removed** from favorites.

TC-06 — Select Currency Pair

Step 14: Click the "ALL" tab in the pair list, find the pair (e.g. `BTC/USDT`), and click it to activate it as the active trading pair. The whole page (order book, ticker, trade form) now shows data for this pair.

TC-02b — Verify 24h Ticker vs Binance

Step 15: Simultaneously fetch the 24h ticker data from the page and from the Binance public API for the same pair, then verify:

- Page last price is a positive number
 - Page last price is within 1% of Binance last price (live price tolerance)
 - Page 24h high matches Binance exactly
 - Page 24h low matches Binance exactly
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TC-03b — Order Book Column Headers

Step 16: Read the order book column headers and verify all three are visible and non-empty: **Price, Amount, Total**.

TC-03c — Order Book View Switches

Step 17: Switch the order book to "Sell-only" view → verify the order book is still visible and the top ask price is positive.

Step 18: Switch to "Buy-only" view → verify the order book is still visible and the top bid price is positive.

Step 19: Switch back to "All" view → verify both top ask and top bid prices are positive.

TC-03d — Order Book Precision Dropdown

Step 20: Cycle through precision values `0.01` → `0.1` → `1` → `0.01` (restoring the default). For each precision, and for each of the three views (all / sell / buy), verify that displayed prices have no more decimal places than the precision allows:

- `0.01` → max 2 decimal places
- `0.1` → max 1 decimal place
- `1` → max 0 decimal places

Restore to "All" view at `0.01` precision at the end.

TC-03e — Order Book LTP & Buy/Sell Ratio (Suggestions Only)

Step 21: Attempt to read the Last Traded Price (LTP) from the order book mid-row. Record as a **suggestion** that LTP display is not yet implemented. No assertion failure.

Step 22: Attempt to read the buy % and sell % ratio bar values. Record as a **suggestion** that the ratio bar is not yet implemented. No assertion failure.

TC-03f — Order Book Has Real Data Rows

Step 23: Count the ask rows and bid rows in the order book, read the top ask price and top bid price. Verify:

- Ask row count > 0
 - Bid row count > 0
 - Top ask price > 0
 - Top bid price > 0
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TC-03g — Valid Spread

Step 24: Read the top bid and top ask prices. Verify that top bid < top ask (a valid market spread exists).

TC-03h — Bid/Ask vs Binance within 0.5%

Step 25: Simultaneously fetch the top bid/ask from the page and from the Binance order book API. Verify:

- Page top bid is within 0.5% of Binance top bid
 - Page top ask is within 0.5% of Binance top ask
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TC-03i — Order Book LTP vs Binance within 0.5%

Step 26: Simultaneously read the page order book LTP and the Binance 24h last price. Verify the LTP is positive and within 0.5% of the Binance last price.

TC-06a — Market Buy Tab Price Field is Disabled

Step 27: Click the **Buy** tab, then click the **Market** tab to activate the Market Buy form.

Step 28: Verify the Price input field is **disabled / read-only** (on Market orders the price is determined by the market, not entered by the user).

TC-06a1 — Buy Button Label

Step 29: Read the text on the Buy button. Verify it contains "BUY" and the base coin name (e.g. "BUY BTC").

TC-06a2 — Quote Currency Available Balance is Non-Negative

Step 30: Read the "Available" balance shown next to the buy form (quote coin, e.g. USDT). Verify it is ≥ 0 .

TC-06a3 — 25% Button

Step 31: Click the **25%** button. Verify the total spend field is filled with a positive value close to 25% of the available USDT balance (within 5% tolerance).

TC-06a4 — 50% Button

Step 32: Click the **50%** button. Verify the total spend field is filled close to 50% of the available USDT balance (within 5% tolerance).

TC-06a5 — 75% Button

Step 33: Click the **75%** button. Verify the total spend field is filled close to 75% of the available USDT balance (within 5% tolerance).

TC-06a6 — 100% Button

Step 34: Click the **100%** button. Verify the total spend field is filled close to 100% of the available USDT balance (within 5% tolerance).

TC-06a7 — Pair Header Shows Selected Pair

Step 35: Read the pair header/title shown above the order form. Verify it contains the base coin name (e.g. "BTC") or the full pair name (e.g. "BTC/USDT").

TC-06b — Capture Pre-Order Balance Snapshot

Step 36: Navigate to the **Portfolio** → **Spot** page and capture a full balance snapshot including:

- Available USDT balance (buy side)
- Available BTC balance (sell side)
- Portfolio spot balances for both USDT and BTC
- Funds balances for both USDT and BTC

Navigate back to the Spot Trading page. Verify the buy available balance is ≥ 0 and matches the portfolio USDT spot balance.

TC-07 — Enter Buy Total and Verify Estimated Fee

Step 37: Click the **Market** tab to ensure it is active.

Step 38: Clear the total field and type the USDT amount to spend (e.g. 8 USDT — value from CSV `buyTotal` column).

Step 39: Read the estimated fee shown on the form. Calculate the expected fee as:

$\text{buyTotal} \times \text{feePercent} \div 100$ (feePercent from CSV, e.g. 0.15%).

Verify the UI estimated fee matches the calculated fee. If the fee element is not displayed, log a suggestion and skip the assertion.

TC-08 — Note USDT Balance Before Placing Order

Step 40: Read and log the current USDT available balance (from the snapshot captured in TC-06b). Verify it is ≥ 0 . This is the baseline for the post-order comparison.

TC-09 — Place Market Buy Order and Verify Success

Step 41: Click the **Market** tab (re-activate to ensure a clean state), then re-enter the buy total amount from CSV.

Step 42: Check if the text "Insufficient balance" is visible on the form within 2 seconds.

- If **YES** → mark order as failed, record a soft failure with the message "Insufficient balance — top up and re-run", and **stop this test**. All remaining tests (TC-09b through TC-14b) will be skipped.

Step 43: If balance is sufficient → record the current timestamp as `orderPlacedAt`.

Step 44: Click the **Buy BTC** button to submit / open the confirmation dialog.

Step 45: In the confirmation dialog/modal, read the executed price, executed amount, and order ID. Click **Confirm** to finalize the order.

Step 46: Wait for the success message/toast. Verify the message contains "success", "created", "placed", or "status:". If it does → `orderSucceeded = true`.

TC-09b — Balance Snapshot After Fill

Skipped if TC-09 failed or balance was insufficient.

Step 47: Wait 2 seconds for the exchange to process the fill.

Step 48: Navigate to Portfolio and capture a new full balance snapshot.

Step 49: Calculate expected balance changes:

- USDT should **decrease** by `buyTotal`
- BTC should **increase** by `executedAmount`

Step 50: Compare the before-snapshot (TC-06b) vs the after-snapshot for all of the following, each within 0.5% tolerance:

- Buy available balance (USDT)
- Sell available balance (BTC)
- Portfolio USDT spot balance
- Portfolio BTC spot balance
- Funds USDT amount
- Funds BTC amount

If a coin is not found in the portfolio, log a suggestion instead of failing.

TC-09c — Market Trades Panel Has Correct Headers

Step 51: Click the **Market Trades** tab in the bottom panel. Read the column headers. Verify they include "price", "amount" (or "qty"), and "time".

TC-09d — My Trades Panel Has Correct Headers

Step 52: Click the **My Trades** tab. Read its column headers. Verify they include "price", "amount" (or "qty"), and "time". Switch back to Market Trades tab.

TC-09e — Market Trades Has Real Data Rows

Step 53: Read the data rows in the Market Trades panel. Verify:

- At least 1 row exists
 - The first row's price is positive
 - The first row's amount is positive
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TC-09f — Market Trades Prices Match Binance within 1%

Step 54: Simultaneously read the Market Trades rows from the page and fetch the 20 most recent trades from the Binance API. Verify the most recent page trade price is within 1% of the Binance most recent trade price.

TC-10 — Market Buy is NOT in Open Orders

Skipped if TC-09 failed.

Step 55: Open the **Open Orders** tab. Look for a row matching the pair. Verify the buy order is **not present as a pending entry** — market orders fill immediately and should never appear in

Open Orders.

TC-11 — All Orders Shows Market Buy as Filled

Skipped if TC-09 failed.

Step 56: Open the **All Orders** tab. Find the latest row. Verify:

- Pair column contains "BTCUSDT" (slash-stripped comparison)
 - Side contains "buy"
 - Type contains "market"
 - Status matches `done` / `filled` / `complete`
 - Executed price > 0
 - Filled amount > 0
 - Remaining = 0
 - Total > 0
 - Date/time is within 60 seconds of when the order was placed (`orderPlacedAt`)
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TC-12 — My Trades Shows the Executed Buy Entry

Skipped if TC-09 failed.

Step 57: Read the **My Trades** tab entries. Verify:

- At least 1 entry exists
 - The entry price is positive
 - The entry amount is positive
 - The entry timestamp is within 2 minutes of the order placement time
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TC-13 — USDT Decreases by Actual Total; BTC Increases by Filled Minus Fee

Skipped if TC-09 failed.

Step 58: Verify the All Orders total does not exceed the entered USDT total by more than 5% (guard against market order overspend bug).

Step 59: Read the current USDT available balance from the trade form. Verify it has decreased by the actual total spent (All Orders total, or entered total if not available), within 3% tolerance.

Step 60: If filled amount is available from All Orders, calculate:

$\text{btcFee} = \text{filledAmount} \times \text{feePercent} \div 100$

$\text{btcReceived} = \text{filledAmount} - \text{btcFee}$

Verify the BTC available balance has increased by `btcReceived`, within 3% tolerance.

TC-14 — Trade History Shows the Market Buy Order

Skipped if TC-09 failed.

Step 61: Open the **Trade History** tab. Find the latest entry matching the order.

- If **no entry found** → log a suggestion ("market order may not appear yet — verify manually") and skip assertions.

Step 62: If entry is found, verify:

- Pair contains "BTCUSDT"
 - Side contains "buy"
 - Executed price > 0
 - Amount > 0
 - Total > 0
 - Fee ≥ 0
 - Date/time is within 60 seconds of the order placement time
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TC-14b — Trade History Bottom Widget Shows the Buy Entry

Skipped if TC-09 failed.

Step 63: Read the first row of the Trade History widget at the bottom of the trading page (this is separate from the full Trade History tab above). Verify:

- Pair references "BTC" or "BTCUSDT"
 - Side contains "buy"
 - Price is positive
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TEARDOWN — After All Tests

Step 64: Close the browser.

Quick Reference Summary

Step Range	Area
1 – 6	Browser launch, login, dismiss dialogs
7	Navigate to Spot Trading page
8	Verify all page labels
9	Search for currency pair
10 – 11	Mark pair as favorite
12 – 13	Unmark pair from favorites
14	Select pair from ALL tab
15	Verify 24h ticker vs Binance
16	Verify order book column headers
17 – 19	Order book view switches (all/sell/buy)
20	Order book precision dropdown
21 – 22	Order book LTP & ratio bar (suggestions)
23 – 24	Order book real rows and valid spread
25 – 26	Order book bid/ask/LTP vs Binance
27 – 28	Market Buy tab Price field is disabled
29	Buy button label is correct
30	Quote currency (USDT) available balance is non-negative
31 – 34	25% / 50% / 75% / 100% percentage buttons
35	Pair header shows selected pair
36	Capture pre-order full balance snapshot
37 – 39	Enter buy total and verify estimated fee
40	Note USDT balance before placing order
41 – 46	Place market buy order and verify success
47 – 50	Post-fill balance snapshot comparison
51	Market Trades panel headers
52	My Trades panel headers

Step Range	Area
53	Market Trades has real data rows
54	Market Trades prices vs Binance within 1%
55	Market buy NOT in Open Orders
56	All Orders shows buy as Filled
57	My Trades shows executed buy entry
58 – 60	USDT decreased and BTC increased after buy
61 – 62	Trade History full tab validation
63	Trade History bottom widget validation
64	Close browser